

Rezwan Hoppe-Islam

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📍 Brookfield, WI

GitHub: <https://github.com/Rez99/data-engineering-portfolio>

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Data Engineer

Data engineer with 15+ years of experience in product analytics, data science, and modern data platforms across e-commerce, healthcare, and technology. Recently completed a production-style data engineering portfolio spanning local lakehouse architecture, cloud infrastructure, and real-time streaming systems. Combines senior analytics judgment, platform thinking, and hands-on engineering to build scalable, reliable, and observable data systems.

Technical Skills

Languages

Python • SQL • R

Streaming & Data Platforms

Kafka • Flink • Spark • DuckDB • PostgreSQL • Iceberg

Transformation & Orchestration

dbt • Airflow

Cloud & Infrastructure

Visualization

Grafana • Superset

Data Engineering Portfolio

Streaming Bot Detection Platform

Designed and implemented a real-time clickstream processing platform using Kafka, Flink, PostgreSQL, and Grafana.

Highlights

- Built an event-driven streaming architecture processing millions of clickstream events.
- Implemented event-time processing, watermarks, and keyed Flink state for live session scoring.
- Designed real-time bot scoring from streaming session statistics and persisted operational scores to PostgreSQL.
- Added replay tooling, schema validation, dead-letter queues, checkpointing, and Grafana observability.
- Diagnosed memory pressure through profiling and reduced TaskManager memory footprint by roughly 50%.
- Investigated Flink execution plans, checkpoint state, JVM memory, and Docker architecture to improve stability.

Technologies

Kafka, Redpanda, Flink, Java, PostgreSQL, Grafana, DuckDB, Parquet, Docker

Cloud Lakehouse

Designed and deployed a cloud-native lakehouse on Google Cloud using infrastructure-as-code.

Highlights

- Provisioned GCP infrastructure with Terraform, including Cloud Storage, Cloud SQL, Cloud Run, IAM, Secret Manager, Workflows, and Dataproc.

- Built Spark and dbt pipelines that load clickstream data into Iceberg tables, create session-level features, train an XGBoost model, and publish evaluation metrics.
- Deployed Polaris and Superset on Cloud Run with Cloud SQL metadata stores.
- Orchestrated the pipeline with Cloud Workflows and ephemeral Dataproc Spark clusters.
- Implemented reproducible setup, execution, dashboard refresh, and teardown workflows.

Technologies

GCP, Terraform, Spark, dbt, Iceberg, Polaris, Cloud Workflows, Cloud Run, Cloud SQL, Dataproc, Superset

Local Lakehouse

Built an end-to-end local lakehouse that transforms 42 million clickstream events into analytical tables, machine-learning features, model predictions, and BI dashboards.

Highlights

- Built Airflow-orchestrated ingestion, transformation, training, and evaluation pipelines.
- Loaded raw clickstream data through staged CSV-to-Parquet-to-Iceberg ingestion.
- Modeled session-level feature stores with dbt and queried analytical datasets with DuckDB.
- Benchmarked Pandas, PostgreSQL, DuckDB, and lakehouse architectures to evaluate storage, memory, and compute tradeoffs.
- Built an external-memory XGBoost purchase prediction model and published model metrics to Superset.

Technologies

DuckDB, dbt, Airflow, Iceberg, Polaris, RustFS, Superset, Python, Parquet, XGBoost

Professional Experience

Senior Data Scientist – Hinge Health

2024–2025

- Led product analytics across Exercise Platform, Exercise Experience, and Computer Vision teams, partnering with 20+ engineers, designers, and PMs.
- Built modular experimentation infrastructure that cut analysis turnaround from hours to minutes and shaped team workflows for StatSig onboarding.

- Developed dbt-powered analytical models to monitor personalization engagement and surface high-friction product moments.
 - Designed KPI frameworks and product health dashboards, including a Computer Vision dashboard that clarified adoption bottlenecks and influenced roadmap focus.
 - Partnered with engineering teams on scalable analytics platforms, experimentation design, and impact measurement.
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Senior Data Scientist – Instacart

2020–2023

- Led experimentation and analytics across Storefront, Cart, Checkout, and Post-Checkout.
 - Delivered insights and A/B test strategy contributing to over \$200M in incremental GMV.
 - Built large-scale experimentation tooling adopted organization-wide and recognized by leadership as a highly impactful data science initiative.
 - Reduced experimentation pipeline runtime by 90%.
 - Reduced infrastructure costs by approximately \$1M annually.
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Consultant – Knowledge Leaps

2019

- Built marketing analytics and forecasting models for grocery retail.
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Data Scientist – Chegg

2017–2018

- Led customer lifetime value and marketing analytics initiatives.
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Product Analyst – eBay / PayPal

2011–2017

- Worked across Checkout, Marketplace, Gift Cards, StubHub, and Growth Experimentation.
 - Selected for eBay Leadership Development Program.
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Analyst – Vodafone

2007–2011

Education

Ph.D. – Biochemical Engineering

University College London

B.Eng. – Biochemical Engineering

University College London